DENIS UZHVA

Machine Learning Engineer | Data Scientist | Open to relocation to Japan | HSP visa-eligible (PhD)

denis.uzhva@yahoo.com   t.me/DenisUzhva   denisuzhva.github.io   github.com/denisuzhva

Profile

PhD machine learning engineer specializing in recommender systems and end-to-end ML pipelines, with **6+ years** of experience spanning industry and research. Currently building and deploying recommendation models (classical and deep learning) for a large-scale music streaming platform, with a track record of shipping measurable improvements to production. Background also includes applied ML research in physics and engineering (CERN, geophysical data compression, multi-agent networks). Actively seeking to relocate to Japan, having already engaged with the local tech community through a Tokyo-based hackathon on ML-driven recommendations.

Skills

- Python, SQL
- DS/DE: PySpark, Polars, Airflow
- ML/DL: Catboost, PyTorch, MLFlow, ClearML, Implicit, Scikit-learn
- NLP: LLM, NLTK
- Mathematics: Statistics, Optimization
- Git, Docker, FastAPI, LaTeX, Claude Code
- Soft skills: business communication, advanced problem solving, team leadership

Education

Ph.D. (Mathematics and Computer Science)

Saint Petersburg State University

2021–2025

M.Sc. (Software Engineering)

Saint Petersburg State University

2019–2021

B.Sc. (Physics)

Saint Petersburg State University

2015–2019

Languages

- Russian: native
- English: proficient
- Japanese: casual

Experience

Machine Learning Engineer

Zvuk LLC

December 2023 – Present

- Developing and supporting **ML/DL recommender services** for audio streaming.
- Full pipeline from experiments to scheduled automatic data preparation, model training (Airflow, MLFlow, ClearML) and service deployment (FastAPI, Ray Serve, Seldon).
- Implementing both classical recommender systems (ALS, boosting) and advanced deep learning (SasRec, Two-Tower) models for **automated music playlist generation**.
- *Key achievements*: developed and deployed a recommender system for **generative playlists** based on pre-defined artists or tracks, which improved skip rate by -4.69%, ATS by +7.16% and coverage by +36.97%.

Machine Learning Engineer, Researcher

Saint Petersburg State University

January 2021 – December 2023

- Developed **deep learning** and noise simulation pipelines for night photography enhancement and geophysical data encoding.
- Developed **reinforcement learning** models for solid body topology prediction, based on desired body durability.
- *Key achievements*: the developed ML/DL systems outperform conventional and state-of-the-art approach by a significant margin; the developed geophysical data compression tool outperforms ZIP and RAR 3 times.

Research Engineer

Institute for Problems in Mechanical Engineering of the RAS

October 2021 – October 2023

- Developed mathematical theory for efficient multi-agent networks control.
- Responsible for the development of the software environment for network simulations.
- Developed **deep learning** models for efficient multi-agent system encoding.
- *Key achievements*: the developed **scalable machine learning systems** successfully encode **big data** emerging in large-scale networks by compressing it approximately 300 times with 0.3% data loss.

Machine Learning Engineer, Research Assistant

Saint Petersburg State University

September 2019 – December 2020

- Implemented state-of-the-art machine learning methods for **data analysis** in the CERN physical experiments.
- Responsible for design of architecture and software implementation of machine learning models, as well as **data augmentation, feature generation and visualization**.
- *Key achievement*: the developed models demonstrate superior accuracy 92.8% in comparison with classical approaches such as decision trees and "cut-based analysis", which achieve 86.7%.

Hackathons and Olympiads

hello, yaponiya!

April 2025 Tokyo, Japan

- Our team has proposed and developed a multi-platform application, where employees receive department recommendations within their company, based on their skills, by a machine learning recommender system.

iVision

2nd place

January 2022 Saint Petersburg, Russia

- Our team has developed a car accident detection system, which predicts accident occurrences by video fragments.
- The proposed final machine learning model predicts car accidents with 70% accuracy, which was the second-best result.

Conferences and Publications

22nd IFAC World Congress

July 2023 Yokohama, Japan

"Adaptive Distributed Cluster Flow Control for a Group of Autonomous Robots"

61st IEEE Conference on Decision and Control

December 2022 Cancún, Mexico

"Compressed Cluster Sensing in Multi-Agent IoT Control"

18th Russian Conference on Artificial Intelligence

October 2020 Moscow, Russia

"Invariance Preserving Control of Clusters Recognized in Networks of Kuramoto Oscillators"

NA61/SHINE CERN Analysis/Software/Calibration Meeting 2019

May 2019 Katowice, Poland

"Convolutional Neural Network for Centrality in Fixed Target Experiments"

Science And Progress

November 2018 Saint Petersburg, Russia

"Investigation of Deep Learning Methods for the Classification of Events in the NOvA Experiment"

Pet Projects

Text improvement engine (NLP) |

October 2023 NLTK BERT

- A tool that analyses a given text and suggests improvements based on the similarity to a list of "standardised" phrases.
- Pre-processes input text with NLTK: make lowercase, lemmatize, expand, etc.
- Considers context with the help of BERT LLM.

Diffusion model for generative music |

November 2022 PyTorch Torchaudio

- Generates raw audio signals in short chunks.
- Preserves context with up to 5 minutes with a WaveNet-like dilated transformer.

Compressed sensing blockchain |

March 2022 NumPy Matplotlib Scikit-learn

- Exploits compressed sensing with measurement matrix sampled from a PRNG with a seed based on previous blocks.
- Resists injection-type attacks.

Image processing Telegram bot |

May 2020 OpenCV Flask DeepAPI

- Telegram bot with an arsenal of image processing techniques.
- Features: color inversion; contrast, brightness and saturation adjustment; AI upscaling and colorization; etc.